Complex Numbers: De Moivre's Theorem - Powers

Questions 1–33

Question 1. Given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Find z^{-1}

b) Hence show that $2\cos\theta = z + z^{-1}$ and $2\mathrm{i}\sin\theta = z - z^{-1}$

Question 2.

a) Find z^{-n}

b) Show that $\cos n\theta = \frac{z^n + z^{-n}}{2}$

c) Show that $\sin n\theta = \frac{z^n - z^{-n}}{2\mathrm{i}}$

Question 3. Find the expansion of each of the following

a) $\left(z+\frac{1}{z}\right)^2$

b) $\left(z+\frac{1}{z}\right)^3$

c) $\left(z+\frac{1}{z}\right)^4$

d) $\left(z+\frac{1}{z}\right)^5$

e) $\left(z+\frac{1}{z}\right)^6$

Question 4. Find the expansion of each of the following

a) $\left(z-\frac{1}{z}\right)^2$

b) $\left(z-\frac{1}{z}\right)^3$

c) $\left(z-\frac{1}{z}\right)^4$

d) $\left(z-\frac{1}{z}\right)^5$

e) $\left(z-\frac{1}{z}\right)^6$

Prerequisite Knowledge for Q5-Q7: Exponential Form

Question 5. Use Euler's formula to write each of the following in terms of $e^{i\theta}$ and $e^{-i\theta}$

a) $\cos\theta$

b) $\sin\theta$

Question 6. Find the expansion of each of the following:

a) $(e^{i\theta} + e^{-i\theta})^2$

b) $(e^{i\theta} + e^{-i\theta})^3$

c) $(e^{i\theta} + e^{-i\theta})^4$

Question 7. Find the expansion of each of the following:

a) $\left[\frac{1}{\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})\right]^2$

b) $\left[\frac{1}{\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})\right]^3$

c) $\left[\frac{1}{\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})\right]^4$

Question 8. Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z + \frac{1}{z}\right)^2$ to show that $\cos^2\theta = \frac{1}{2}\cos 2\theta + \frac{1}{2}$

Question 9. Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z - \frac{1}{z}\right)^2$ to show that $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$

Prerequisite knowledge for Q10: Exponential Form and Integration of Standard Functions

Question 10.

a) By writing $\cos\theta$ in terms of $\mathrm{e}^{\mathrm{i}\theta}$ and $\mathrm{e}^{-\mathrm{i}\theta}$ show that $\cos^3\theta = \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta$

b) Hence show that

$$\int_0^{\frac{\pi}{3}} \cos^3\theta \,\mathrm{d}\theta = \frac{p}{q}\sqrt{3}$$

where p and q are constants to be found.

Question 11.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z - \frac{1}{z}\right)^3$ to show that $\sin^3\theta = \frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta$

b) Hence solve $\frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta = \frac{1}{8}$ in the interval $0 \leq \theta \leq 2\pi$

Question 12.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z - \frac{1}{z}\right)^4$ to show that

$$\sin^4\theta = \frac{1}{8}(a\cos 4\theta + b\cos 2\theta + c)$$

where a, b and c are constants to be found.

b) Hence show that

$$\sin\frac{\pi}{8} = \left(\frac{3 - 2\sqrt{2}}{8}\right)^{\frac{1}{4}}$$

Prerequisite knowledge for Q13: Reversing the Chain Rule

Question 13.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z + \frac{1}{z}\right)^4$ to find $\cos^4\theta$ in terms of cosines of multiple angles.

b) Hence show that

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 8\cos^4\theta \,\mathrm{d}\theta = 3\pi$$

Prerequisite Knowledge for Q14-Q15: Exponential Form

Question 14. Will is proving that

$$32\sin^4\theta \equiv 4\cos 4\theta - 16\cos 2\theta + 12$$

His workings are shown below:

$$\begin{aligned}
16\sin^4\theta &\equiv (2\sin\theta)^4 \\
&\equiv (\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})^4 \\
&\equiv 1(\mathrm{e}^{\mathrm{i}\theta})^4 + 4(\mathrm{e}^{\mathrm{i}\theta})^3(\mathrm{e}^{-\mathrm{i}\theta}) + 6(\mathrm{e}^{\mathrm{i}\theta})^2(\mathrm{e}^{-\mathrm{i}\theta})^2 \\
&\quad + 4(\mathrm{e}^{\mathrm{i}\theta})(\mathrm{e}^{-\mathrm{i}\theta})^3 + (\mathrm{e}^{-\mathrm{i}\theta})^4 \\
&\equiv \mathrm{e}^{4\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta} + 4\mathrm{e}^{2\mathrm{i}\theta} + 4\mathrm{e}^{-\mathrm{i}2\theta} + 6 \\
&\equiv 2\sin 4\theta + 4\sin 2\theta + 6
\end{aligned}$$

So that

$$32\sin^4 x \equiv 4\sin 4\theta + 8\sin 2\theta + 12$$

a) Find the error in his working.

b) Fully correct his working.

c) Use a similar method to show that

$$32\cos^4\theta \equiv 4\cos 4\theta + 16\cos 2\theta + 12$$

Question 15.

a) By writing $\cos\theta$ in terms of $\mathrm{e}^{\mathrm{i}\theta}$ and $\mathrm{e}^{-\mathrm{i}\theta}$ find $\cos^5\theta$ in terms of cosines of multiple angles.

b) Hence solve

$$2\cos 5\theta + 10\cos 3\theta + 20\cos\theta - 1 = 0$$

in the interval $0 \leq \theta \leq 2\pi$ giving your answers in exact form.

Prerequisite knowledge for Q16: Reversing the Chain Rule

Question 16.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z - \frac{1}{z}\right)^5$ to find $\sin^5\theta$ in terms of cosines of multiple angles.

b) Hence show that

$$\int_0^{\frac{\pi}{2}} 16\sin^5\theta\,\mathrm{d}\theta = \frac{128}{15}$$

Question 17.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z + \frac{1}{z}\right)^6$ to find $\cos^6\theta$ in terms of cosines of multiple angles.

b) Hence show that

$$\cos\frac{\pi}{12} = \frac{1}{2}(26 + 15\sqrt{3})^{\frac{1}{6}}$$

Prerequisite knowledge for Q18: Exponential Form

Question 18.

a) By writing $\sin\theta$ in terms of $\mathrm{e}^{\mathrm{i}\theta}$ and $\mathrm{e}^{-\mathrm{i}\theta}$ show that

$$32\sin^6\theta = -\cos 6\theta + a\cos 4\theta - b\cos 2\theta + c$$

where a, b and c are positive constants to find.

b) Hence show that

$$\sin\frac{\pi}{12} = \frac{1}{2}(26 - 15\sqrt{3})^{\frac{1}{6}}$$

Question 19. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$. By expanding $\left(z^3 + \frac{1}{z^3}\right)^2$, show that

$$2\cos^2 3\theta = 1 + \cos 6\theta$$

Question 20. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$. By expanding $\left(z^4 - \frac{1}{z^4}\right)^3$, show that

$$4\sin^3 4\theta = 3\sin 4\theta - \sin 12\theta$$

Question 21. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$. By expanding $\left(z^3 - \frac{1}{z^3}\right)^5$, to find an expression for $\sin^5 3\theta$ in terms of sines of multiple angles.

Question 22. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Find $z^n + \frac{1}{z^n}$ in trigonometric form.

b) Find $z^n - \frac{1}{z^n}$ in trigonometric form.

c) Hence, using $\left(z + \frac{1}{z}\right)^2\left(z - \frac{1}{z}\right)^2$ show that

$$\sin^2\theta\cos^2\theta = \frac{1}{8}(1 - \cos 4\theta)$$

Question 23. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Find $z^n + \frac{1}{z^n}$ in trigonometric form.

b) Find $z^n - \frac{1}{z^n}$ in trigonometric form.

c) Hence, using $\left(z + \frac{1}{z}\right)^3\left(z - \frac{1}{z}\right)^3$ show that

$$\sin^3\theta\cos^3\theta = \frac{1}{32}(3\sin 2\theta - \sin 6\theta)$$

Question 24. Without using a trigonometric identity, show that

$$\cos^3\theta\sin^4\theta = \frac{1}{64}(a\cos 7\theta + b\cos 5\theta + c\cos 3\theta + d\cos\theta)$$

where a, b, c and d are integers to be found.

Prerequisite knowledge for Q25-Q26: Exponential Form

Question 25.

a) Show that

$$\mathrm{Re}\left(\mathrm{e}^{3\mathrm{i}\theta}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})^3\right) = k\cos 3\theta\cos^3\theta$$

where k is an integer to be found.

b) Hence show that

$$\cos\frac{\pi}{12} = \sqrt[3]{\frac{p\sqrt{2}+q\sqrt{6}}{16}}$$

where p and q are integers to be found.

Question 26.

a) Show that

$$\mathrm{Im}\left(\mathrm{e}^{4\mathrm{i}\theta}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})^4\right) = k\sin 4\theta\sin^4\theta$$

where k is an integer to be found.

b) Hence show that

$$\sin\frac{\pi}{8} = \frac{1}{2}(p-q\sqrt{2})^{\frac{1}{4}}$$

where p and q are integers to be found.

Prerequisite knowledge for Q27-Q29: Integration by Substitution and/or Reversing the Chain Rule

Question 27. Find $\int \cos^4\theta \sin\theta \, \mathrm{d}\theta$:

a) By using standard integration methods.

b) By first writing $\cos^4\theta \sin\theta$ in terms of sines of multiple angles.

Question 28.

a) Show that

$$16\sin^3\theta\cos^2\theta = a\sin\theta + b\sin 3\theta + c\sin 5\theta$$

where a, b and c are constants to be found.

b) Hence, by using $x = \sin\theta$ show that

$$\int_0^1 16x^3\sqrt{1-x^2}\,\mathrm{d}x = \frac{32}{15}$$

Question 29.

a) Show that

$$8\sin^2\theta\cos^2\theta = a - b\cos 4\theta$$

where a and b are constants to be found.

b) Hence, by using a suitable substitution show that

$$\int_0^2 4x^2\sqrt{4-x^2}\,\mathrm{d}x = 4\pi$$

Prerequisite knowledge for Q30: Reduction Formulae for Integration

Question 30.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z - \frac{1}{z}\right)^6$ to write $\sin^6\theta$ in terms of cosines of multiple angles.

b) Hence show that

$$\int_0^{\frac{\pi}{4}} \sin^6\theta\,\mathrm{d}\theta = \frac{p\pi - q}{r}$$

where p, q and r are constants to be found.

c) Verify your result in part b) by using the formula

$$I_n = \left(\frac{n-1}{n}\right) I_{n-2} - \frac{1}{n2^{\frac{n}{2}}}$$

where $I_n = \int_0^{\frac{\pi}{4}} \sin^n\theta\,\mathrm{d}\theta$

Question 31. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Show that $z^n + \frac{1}{z^n} = 2\cos n\theta$

b) Show that $(2x-1)$ is a factor of $8x^3 - 12x^2 + 6x - 1$

c) Use $\left(z + \frac{1}{z}\right)^3$ to show that

$$8\cos^3\theta = 6\cos\theta + 2\cos 3\theta$$

d) Hence, by considering $\left(z + \frac{1}{z}\right)^3$, $\left(z + \frac{1}{z}\right)^2$ and $z + \frac{1}{z}$ solve

$$z^6 - 3z^5 + 6z^4 - 7z^3 + 6z^2 - 3z + 1 = 0$$

giving your answers in the form $a + b\mathrm{i}$ where a and b are constants to be found.

Question 32. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Show that $z^n + \frac{1}{z^n} = 2\cos n\theta$

b) Use $\left(z + \frac{1}{z}\right)^3$ to show that

$$\cos^3\theta = \frac{1}{4}(3\cos\theta + \cos 3\theta)$$

c) It is given that

$$f(z) = z^6 + 3z^4 + z^3 + 3z^2 + 1$$

Considering $f(z) = 0$, write $f(z)$ as a product of a quadratic and a quartic factor.

Question 33. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Use $\left(z + \frac{1}{z}\right)^5$ to show that

$$32\cos^5\theta = 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta$$

b) Hence write

$$z^{10} + 5z^8 + 10z^6 - z^5 + 10z^4 + 5z^2 + 1$$

as a product of two polynomials.